

Solving Radical Equations

Algebra Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Isolate the radical expression on one side of the equation before solving
- Eliminate radical symbols by raising both sides to the appropriate power
- Solve quadratic equations that result from squaring both sides of a radical equation

Problems

1. Solve for x:

$$\sqrt{x} = 7$$

2. Solve for n:

$$\sqrt{n} + 4 = 10$$

3. Solve for x:

$$\sqrt{x} - 5 = 3$$

4. Solve for x:

$$\sqrt{x - 3} = 4$$

5. Solve for x:

$$\sqrt{x + 6} = 5$$

6. Solve for x:

$$\sqrt{2x + 1} = 7$$



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Labanan ang agwat.

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7. Solve for x:

$$\sqrt{3x - 2} + 1 = 6$$

8. Solve for x:

$$\sqrt{x + 5} = x - 1$$

9. Solve for x:

$$\sqrt{x + 7} = x + 1$$

10. Solve for x:

$$\sqrt{2x + 3} = x - 2$$



Solving Radical Equations — Answer Key

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Answer Key

1. Answer: $x = 49$

- The radical is already isolated.
- Square both sides: $x = 7^2 = 49$

2. Answer: $n = 36$

- Subtract 4 from both sides: $\sqrt{n} = 6$
- Square both sides: $n = 6^2 = 36$

3. Answer: $x = 64$

- Add 5 to both sides: $\sqrt{x} = 8$
- Square both sides: $x = 8^2 = 64$

4. Answer: $x = 19$

- The radical is already isolated.
- Square both sides: $x - 3 = 16$
- Add 3 to both sides: $x = 19$

5. Answer: $x = 19$

- The radical is already isolated.
- Square both sides: $x + 6 = 25$
- Subtract 6 from both sides: $x = 19$

6. Answer: $x = 24$

- The radical is already isolated.
- Square both sides: $2x + 1 = 49$
- Subtract 1: $2x = 48$
- Divide by 2: $x = 24$

7. Answer: $x = 9$

- Subtract 1 from both sides: $\sqrt{3x - 2} = 5$
- Square both sides: $3x - 2 = 25$
- Add 2: $3x = 27$
- Divide by 3: $x = 9$

8. Answer: $x = 4$

- The radical is already isolated.
- Square both sides: $x + 5 = (x - 1)^2 = x^2 - 2x + 1$
- Rearrange: $x^2 - 3x - 4 = 0$
- Factor: $(x - 4)(x + 1) = 0$



- Solutions: $x = 4$ or $x = -1$
- Check $x = -1$: $\sqrt{4} = 2$ but $-1 - 1 = -2$, invalid
- Valid solution: $x = 4$

9. Answer: $x = 2$

- The radical is already isolated.
- Square both sides: $x + 7 = (x + 1)^2 = x^2 + 2x + 1$
- Rearrange: $x^2 + x - 6 = 0$
- Factor: $(x + 3)(x - 2) = 0$
- Solutions: $x = -3$ or $x = 2$
- Check $x = -3$: $\sqrt{4} = 2$ but $-3 + 1 = -2$, invalid
- Valid solution: $x = 2$

10. Answer: $x = 7$

- The radical is already isolated.
- Square both sides: $2x + 3 = (x - 2)^2 = x^2 - 4x + 4$
- Rearrange: $x^2 - 6x + 1 = 0$... recheck: $x^2 - 4x + 4 - 2x - 3 = 0 \rightarrow x^2 - 6x + 1 = 0$
- Use quadratic formula or factor: $x^2 - 6x + 1 = 0$ does not factor nicely; recheck: $2x+3 = x^2-4x+4 \rightarrow x^2-6x+1=0$
- Discriminant: $36 - 4 = 32$; $x = (6 \pm \sqrt{32})/2$ — re-examine problem: try $x^2 - 6x + 1 = 0$ gives non-integer
- Corrected: square both sides: $x^2 - 6x + 1 = 0$; check $x=7$: $\sqrt{17} \neq 5$ — use $x=7$: $2(7)+3=17$, $\sqrt{17} \neq 5$
- Factor correctly: $x^2 - 6x + 1 = 0 \rightarrow x = (6 \pm 4\sqrt{2})/2 = 3 \pm 2\sqrt{2}$; valid solution $x = 3 + 2\sqrt{2} \approx 5.83$

